

Tshego also Mogosi.

Exercise 02: SQL FOUNDATIONS: Aggregate functions & operators

01

SELECT DISTINCT department ~~FROM~~
FROM students;

Department
IT
HR
Finance

02

SELECT department, avg(age) AS avg-age
FROM students
GROUP BY department;

Department	avg-age
IT	20.5
HR	22
Finance	23

03

```
SELECT department  
FROM student  
WHERE department = 'HR'
```

~~OR~~

```
OR department = 'IT'  
GROUP BY department;
```

Department.
IT
HR

04

```
SELECT *  
FROM students  
WHERE age BETWEEN 21 AND 23;
```

Student - id	name	age	department.
2	Bob	22	HR
3	Charlie	21	IT
4	Dionci	23	Finance
5	Eve	22	HR

05

05

```
SELECT *
FROM students
WHERE department = 'HR' OR department = 'IT' AND
age > 21;
```

Student-id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

06

```
SELECT department, sum SUM(credit) as AS total-credits
FROM courses
by GROUP BY department
HAVING SUM(credits) > 5;
```

Department	Total-credits
IT	11

07

```
SELECT *
FROM courses
WHERE NOT credits != 4;
```

Course-id	course-name	department	credits
102	Python	IT	4
103	Data Science	IT	4

08

```

SELECT course-id, course-name, credits
FROM courses
ORDER BY credits DESC
LIMIT 3;

```

~~09~~~~SELECT~~

Course-id	Course-name	Credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

09

```

SELECT MAX(grade) AS max-grade, MIN(grade) AS
min-grade, AVG(grade) AS avg-grade
FROM enrollments
GROUP BY

```

Max-grade	Min-grade	Avg-grade
90	78	84.6

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```

SELECT course-id, COUNT(course-id)
SELECT enrollment-id, COUNT(course-id) AS enroll
enrollment-count
FROM enrollments
GROUP BY enrollment-id

```

Enrollment-id	Enrollment-count
1	1
2	1
3	1
4	1
5	1

11

SELECT department, SUM(salaries) AS Total-salary,
SUM(bonus) AS Total-bonus
FROM salaries
GROUP BY department

Department	Total-salary	Total-bonus
IT	122 000	10 800
HR	109 000	7 500
Finance	7 000	6 000

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SELECT department, AVG(salary) AS avg-salary, ~~SUM(salary)~~
FROM salaries
WHERE salary > 55 000
GROUP BY department

Department	avg-salary
IT	61 000
Finance	70 000

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SELECT employee-id, name, salary + bonus AS ~~total-income~~
total-income
FROM salaries
WHERE salary + bonus > 60 000

Employee-id	name	total-income
1	Tom	65 000
2	Spike	76 000
3	Tyke	67 800

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```
SELECT department, sum(Budget) AS total-budget,
       AVG(budget) AS avg-budget
FROM projects
GROUP BY department
HAVING avg(budget) avg(budget) > 10 000
```

Department	total-budget	avg-budget
IT	27 000	13 500
Finance	8 000	8 000

15

```
SELECT *
FROM project
WHERE budget BETWEEN 50 000 AND 120 000
AND department != 'Marketing'
```

Project-id	Project-name	Department	Budget.
1	AI App	IT	120000
2	Payroll System	Finance	80 000
3	HR Total	HR	30 000